one point or another in the vicinity of the nucleus, a *probability value* for the distribution of electrons. But all these photographic records as a group—being collected in a single instant, as it were—can be regarded also as a picture of the average distribution of the density of electricity in the vicinity of the nucleus. Such a density distribution is comparable, in some degree, to a wave phenomenon—a stationary wave. If we conceive of stationary waves of electrically negatively charged matter, these, too, will correspond to a certain specific density distribution of this matter. Actually, the de Broglie matter waves can be interpreted so that the square of the amplitude of the waves at a certain specific point indicates the density of matter at that spot. But we can just as well say that it indicates the possibility that in a snapshot the electron will be discovered at that very point. These stationary waves in the immediate vicinity of the nucleus were investigated by Schrödinger; they represent, in fact, the object of his wave mechanics. These stationary waves also form the foundation for verifying a stationary distribution of electricity in the vicinity of the nucleus, and consequently, the existence of stationary states of the atom, in which no radiation occurs, becomes understandable. Thus one of the difficulties in connection with the atom model is eliminated.

The individual stationary waves which can possibly occur in an atom, do not form a continuous series, any more than do the individual characteristic vibrations of the strings of a musical instrument. A string can vibrate in its fundamental tone, in which case it has no nodes. On the other hand, it can vibrate also in an overtone, and in that case it has one or more nodes. Similarly, an atom can ‘vibrate’ in its ground state, in which case it has no ‘nodes’—meaning, no levels on which the density of matter disappears. But it can vibrate also in an overtone—in an ‘excited’ vibration, as it is called—and then there are several nodal levels of zero density. These various stationary vibrations correspond to the various stationary states which an atom is capable of assuming.

Let us elucidate the conditions described above by the example of the simplest of atoms, the hydrogen atom, with the aid of illustrations. The ground state of a hydrogen atom (denoted by the symbol Is) was described by Bohr in 1913 as a circular orbit of the electron around the nucleus (Figure 4). In the particle aspect, this is a clear picture. According to this view, the electron has a spin moment about the nucleus. To-day we know that actually it has none. Therefore, to-day we say rather—still within the framework of the particle aspect—that the electron shuttles on a straight line about the nucleus. Thus, we imagine the atoms as shown in Figure 7 (*a*).

But on the other hand, we can conceive of the state of electronic matter in a hydrogen atom also as a wave process (Figure 7 (*b*)). If we should take thousands of snapshots of the atom in its ground state and then develop them superimposed on each other, we would get a density distribution, or